

Practice Problem Sheet-1

Course Code: PHY 2105/PHY 105 Summer: 2026 [262]

Course Title: Physics Content: SHM

1. A mass-spring system with a mass of $m = 80 \text{ g}$ is oscillating in Simple Harmonic Motion (SHM). At equilibrium position, the velocity is 70 ms^{-1} and at amplitude position, the acceleration is 720 ms^{-2} . If we have released the mass from amplitude at $t = 0$, Determine the value of (i) displacement at $t = 3 \text{ s}$, (ii) acceleration at $t = 4 \text{ s}$, and (iii) total energy at $t = 5 \text{ s}$.
2. The equation of displacement of a mass-spring system is, $y = 2\cos(20t+2\pi) \text{ m}$. Calculate (i) Potential Energy at $t = 0.3 \text{ s}$, kinetic energy at $t = 2 \text{ s}$, and (ii) Maximum velocity and maximum acceleration. Given, the mass attached to the spring is 50 g .
3. A body of mass 70 g is attached with a spring of spring constant 18.6 N/m . The body is displaced by 20 cm from its equilibrium position and released at $t = 0$. Then the body executes SHM. Calculate (i) angular frequency, (ii) velocity at $t = 3 \text{ s}$, and (iii) acceleration at $t = 1.5 \text{ s}$.
4. A 0.24 kg body undergoes SHM of amplitude 6 cm and period 0.30 s . (i) What is the magnitude of the maximum force acting on it? (ii) If the oscillations are produced by a spring, what is the spring constant?
5. The equation of displacement of a mass-spring system is, $y = 3\cos(2\pi t - \frac{\pi}{3}) \text{ m}$. Calculate (i) the potential energy and kinetic energy at $t = 2 \text{ s}$, (ii) compare the potential and kinetic energy, (iii) at which time the potential energy will be equal to the kinetic energy? Here the mass of the mass-spring system is 0.45 kg .
6. In an electric shaver, the blade moves back and forth over a distance of 8 mm in SHM, with a frequency 50 Hz . Find (i) the amplitude, (ii) the maximum blade speed, and (iii) the magnitude of the maximum acceleration of the blade.
7. A body of mass 38 gm is attached with a spring of spring constant 740 dynes/cm . The body is displaced by 15 cm from its equilibrium position and released. Then the body executes SHM. Calculate (i) the time period, (ii) frequency, (iii) angular frequency, (iv) maximum velocity, and (v) maximum acceleration.
8. A particle executes SHM about the point $x = 0$. At $t = 0$ it has displacement $x = 0.40 \text{ cm}$ and zero velocity. The frequency of the motion is 0.5 Hz . Determine, (i) the period, (ii) the angular frequency, (iii) the amplitude, (iv) the displacement at $t = 3.0 \text{ s}$, and (v) the velocity at $t = 3.0 \text{ s}$.
9. An object undergoing simple harmonic motion takes 0.25 s to travel from one point of zero velocity to the next such point. The distance between those points is 34 cm . Calculate the (i) period, (ii) frequency, and (iii) amplitude of the motion.
10. A body executes SHM such that its velocity at mean position is 1 m/s and acceleration at one extremity is 1.57 m/s^2 . Calculate time period and angular frequency of oscillation.

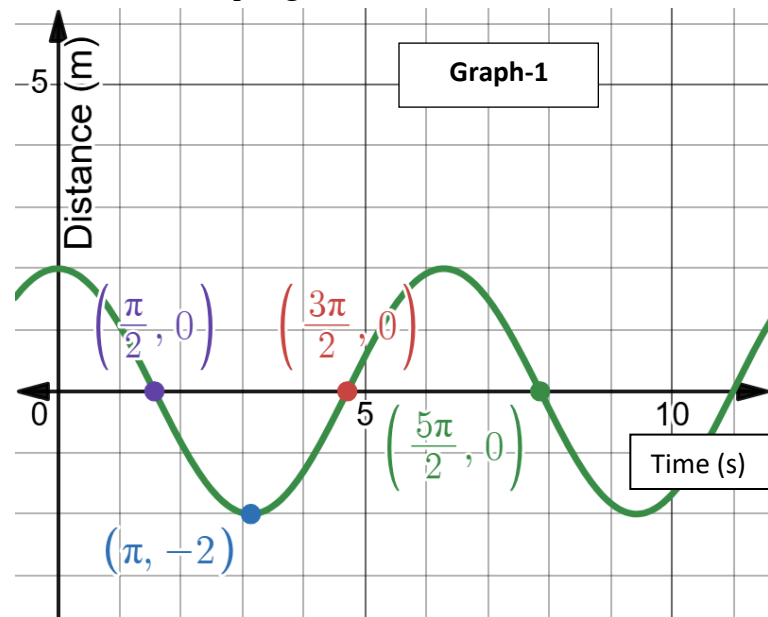
11. Draw the phase difference diagram for the waves: (i) $y_2 = A \sin(\omega t - \frac{\pi}{3})$, $y_1 = A \sin \omega t$; (ii) $y_2 = A \sin(\omega t + \frac{\pi}{4})$, $y_1 = A \sin \omega t$; (iii) $y_1 = A \sin(\omega t + \frac{2\pi}{3})$, $y_2 = A \sin(\omega t + \frac{\pi}{3})$; (iv) $y_1 = A \sin(\omega t + \frac{3\pi}{4})$, $y_2 = A \sin(\omega t - \frac{\pi}{4})$; and (v) $y_1 = A \sin(\omega t - \frac{\pi}{3})$, $y_2 = A \sin(\omega t + \frac{\pi}{3})$.

12. A block whose mass m is 680 gm is fastened to a spring whose spring constant k is 65 N/m. The block is pulled maximum 11 cm generating cosine form on a frictionless surface and released from rest at $t=0$. Find out (i) time period, (ii) angular frequency, (iii) phase constant, (iv) V ($t=4$ sec), (v) $V_{\max}$, (vi) a ($t=4$ sec), (vii) $a_{\max}$, (viii) displacement at $t=0$ sec, $t=7$ sec, and (ix) velocity at a displacement $x=0.11$ m, $x=0.04$ m.

13. A mass oscillates with amplitude of 4.00 m, a frequency of 0.5 Hz and a phase angle of $\pi/4$.
(i) What is the period T ? (ii) Write an equation for the displacement of the particle, and (iii) Calculate the velocity and acceleration of the object at time $t=5$ s.

14. A 2.00 kg block is attached to a spring and force constant of the spring is $k=196$ N/m. The block is held a distance of 5.00 cm from equilibrium and released at $t=0$. (i) Find the angular frequency ω , the frequency f , and the period T . (ii) Write an equation for displacement, x vs. time, t .

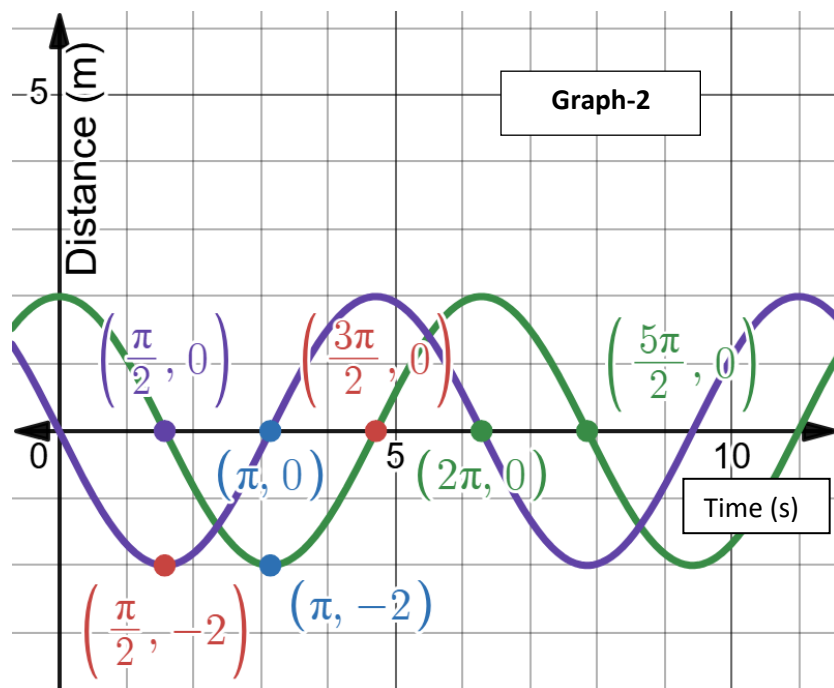
15. Graph-1 illustrates the position vs. displacement graph for a mass spring system oscillating in Simple Harmonic Oscillation. The mass of the mass-spring system is 500 g.
(i) Calculate the equation of displacement vs. time.
(ii) Calculate the acceleration at $t=2.5$ sec.
(iii) Determine the kinetic energy at $t=1.5$ sec.



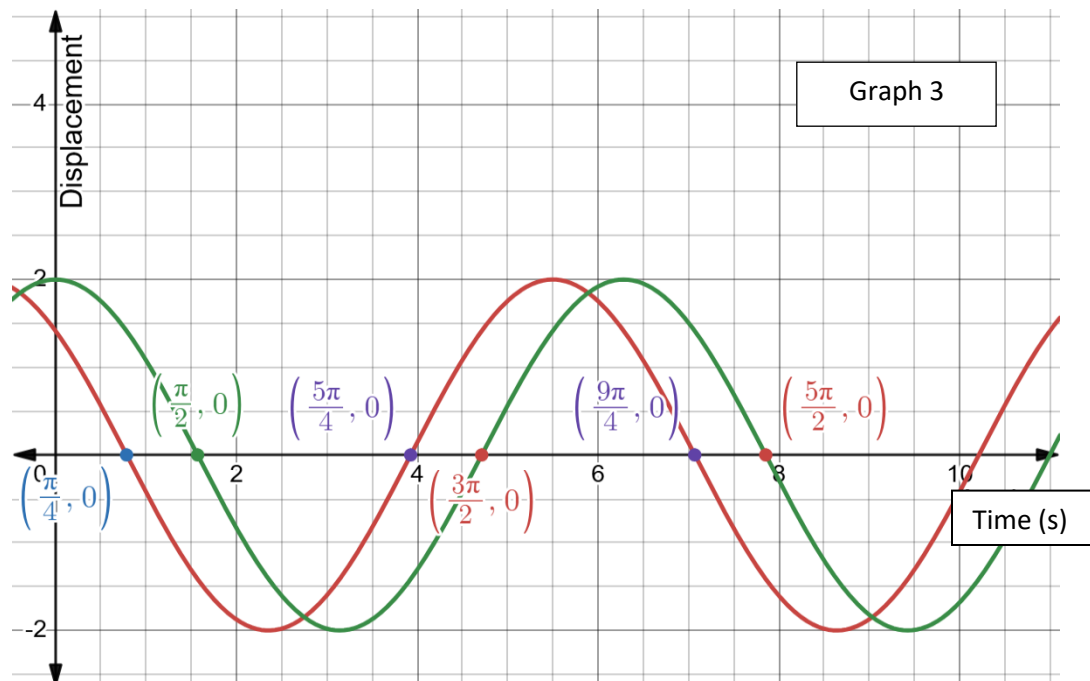
16. Construct the equation of displacement of two objects oscillating in simple harmonic oscillations and find out their phase difference according to Graph-2.

17. Determine the length and frequency of a simple pendulum that will swing back and forth in SHM with a period of 3.5 s.

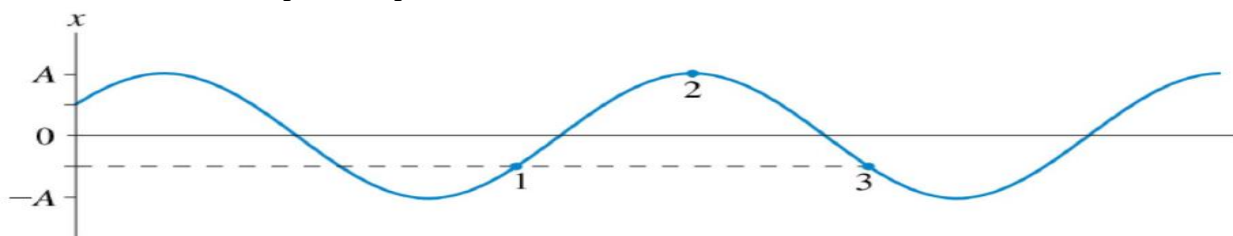
18. An oscillator consists of a block attached to a spring ($k=360$ N/m). At some time t , the position (measured from the system's equilibrium location), velocity, and acceleration of the block are $x=0.150$ m, $v=-15$ m/s, and $a=-125$ m/s². Calculate (i) the frequency of oscillation, (ii) the mass of the block, and (iii) the amplitude of the motion.



19. Construct the equation of displacement of two objects oscillating in simple harmonic oscillations and find out their phase difference according to Graph 3.

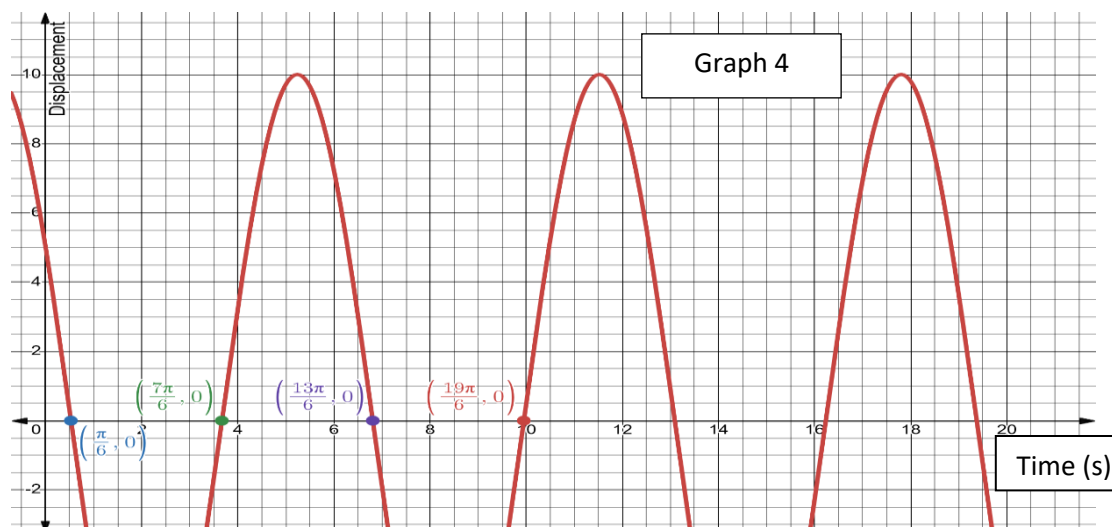


20. A particle executes SHM of amplitude 10 m when the particle is 2.5 m from its mean position, its acceleration is found to be 80 m/s^2 . Find the (i) velocity (ii) time period and (iii) maximum velocity.
21. The figure shows below a position-versus-time graph for a particle in SHM. What is the phase constant? What is the phase at point 2?



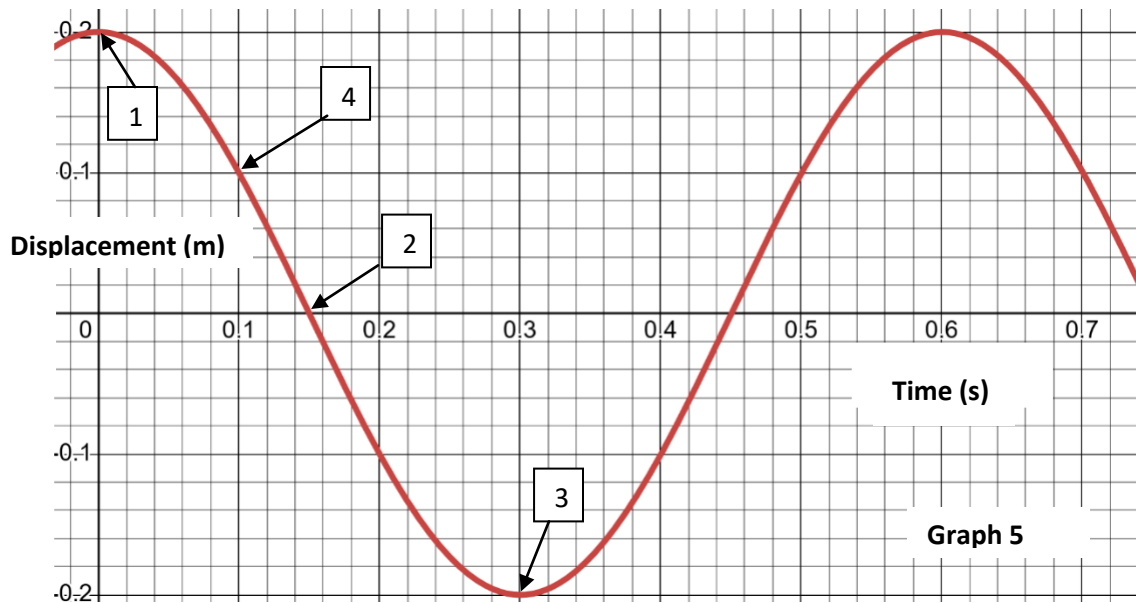
22. An object in simple harmonic motion has amplitude of 4.0 cm, a frequency of 2.0 Hz, and a phase constant of $2\pi/3$ rad. Draw a position graph showing two cycles of motion.
23. Graph 4 illustrates the position vs. displacement graph for a mass spring system oscillating in Simple Harmonic Oscillation. The mass of the mass-spring system is 400 g. (i) Calculate the equation of displacement vs. time, (ii) Calculate the acceleration at $t = 1$ sec, and (iii) Determine the kinetic energy at $t = 0.5$ sec.

24. Suppose a spring block-system moves between top and bottom point of a tall buildings as a mass dampers. The block has mass $m = 2.72 \times 10^5 \text{ kg}$ and



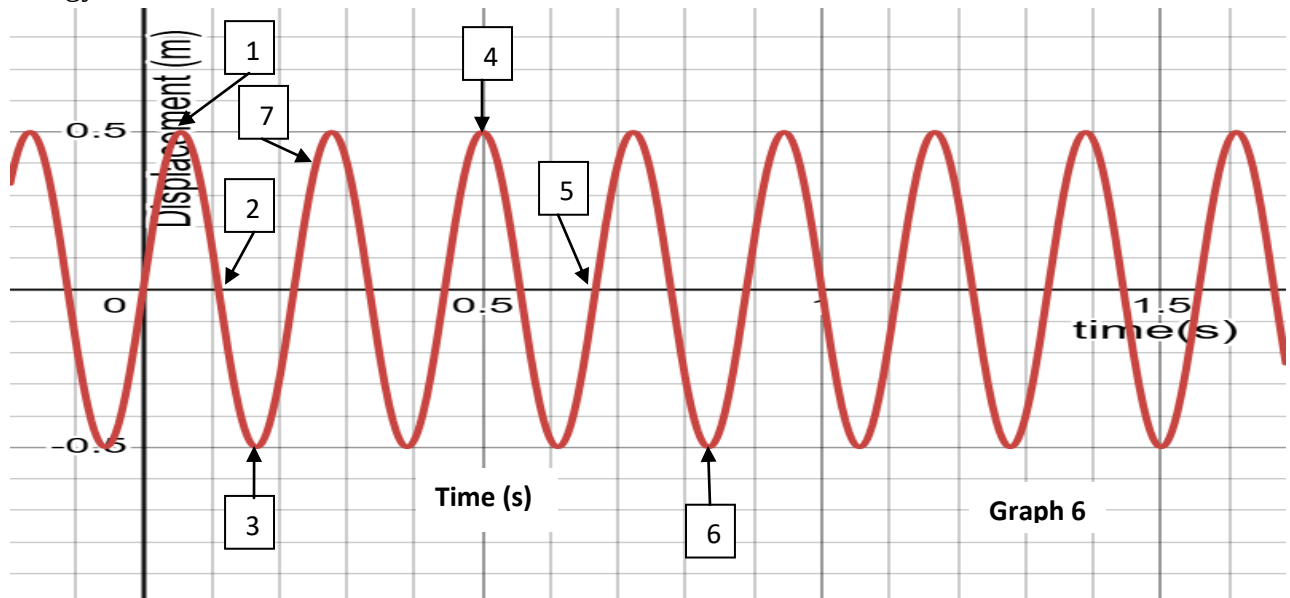
designed to oscillate at a frequency $f=10$ Hz with amplitude $x_m=20$ cm. Calculate (i) the mechanical energy, (ii) what is the block speed as it passes through the equilibrium point?

25. A body of mass 500 gm is suspended from a spring of negligible mass and it stretches the spring by 7 cm. For a displacement of 3 cm it is given a downward velocity 40 cm/s. Calculate (i) the spring constant, (ii) the angular frequency, (iii) the time period (iv) the initial potential energy, (v) the initial kinetic energy, and (vi) the amplitude of the ensuing motion of the spring.
26. Graph 5 illustrates the position vs. displacement graph for a mass spring system oscillating in Simple Harmonic Oscillation. The mass of the mass-spring system is 0.25 kg.

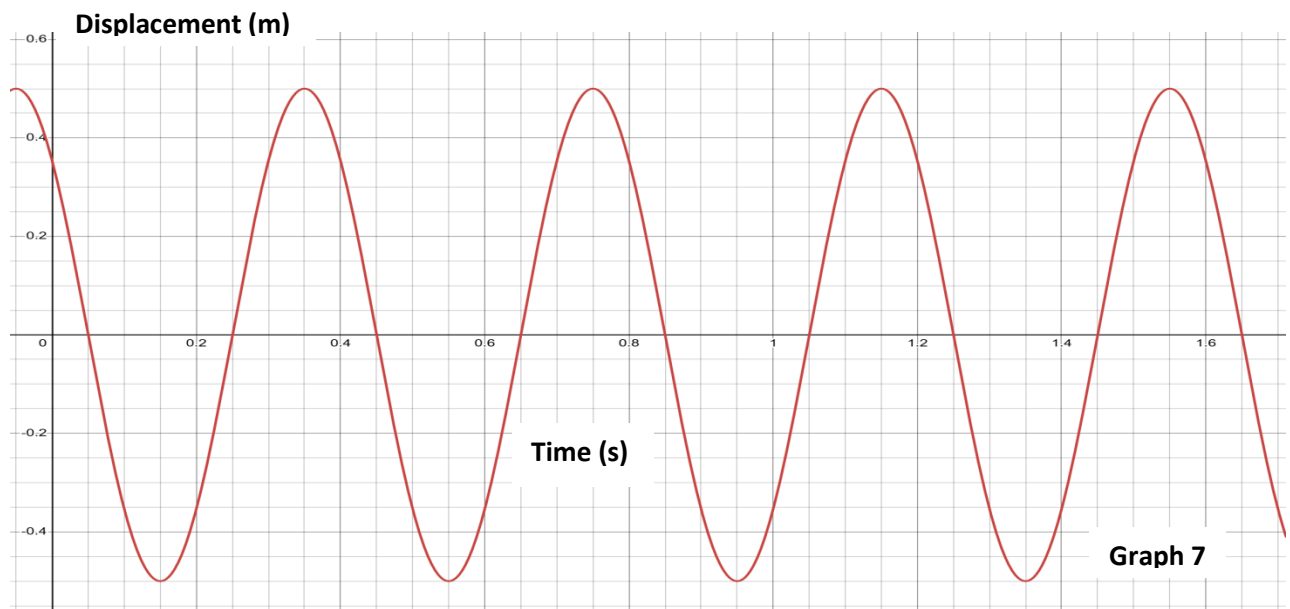


- (i) Find out the equation of displacement vs. time, (ii) Calculate the displacement for $t=0.45$ sec, (iii) Calculate the velocity at $t = 2$ sec, (iv) Calculate the acceleration at $t = 3$ sec, (v) Determine the potential energy at $t = 0.25$ sec, (vi) Determine the kinetic energy at $t = 0.5$ sec, (vii) Determine the total energy at $t=0.6$ sec, (viii) Determine the total energy at $x=7$ cm, and (ix) Determine the total energy of the oscillation, (x) Determine the kinetic energy at $x = 14$ cm from the equilibrium point, (xi) Mathematically calculate the position where the kinetic energy is 0, (xii) Determine the potential energy at $x = 10$ cm from the equilibrium point, (xiii) Determine the position where kinetic energy equals to potential energy, (xiv) Determine the position where potential energy is one quarter of the total energy, and (xv) Determine the position where kinetic energy is half of the total energy, (xvi) $V_{\max}$, $a_{\max}$ at point 3, (xvii) $V_{\max}$, $a_{\max}$ at point 2, (xviii) $V_{\max}$, $a_{\max}$ at point 1, (xix) $V_{\max}$, $a_{\max}$ of the oscillation, and (xx) E_p , E_k and E_T at point 4.
27. At $t=0$, the displacement $x(0)$ of the block in a linear oscillator is -8.50 cm (Read $x(0)$ as “ x at time zero.”). The block’s velocity $v(0)$ then is 0.920 m/s, and its acceleration $a(0)$ is 47.0 m/s^2 . (i) What is the angular frequency ω of this system? (ii) What are the phase constant ϕ and amplitude x_m ?
28. Construct the equation of displacement vs. time from the Graph 6. The mass of the mass-spring system is 200g. Calculate the (i) angular frequency, frequency, time period, and amplitude, (ii) displacement for $t=0.3$ sec, (iii) construct the equation for velocity, v vs. time, t , and velocity at $t = 3$ sec, (iv) construct the equation for acceleration, a vs. time, t , and acceleration at $t = 0.4$ sec, (v) potential energy at $t = 0.15$ sec, (vi) kinetic energy at $t = 0.15$ sec, (vii) total energy, (viii) $V_{\max}$ and $a_{\max}$ at point 4, (ix) v at point 2 and a at point 6, (x) $V_{\max}$, $a_{\max}$, and E_T at point 5, (xi) $V_{\max}$, $a_{\max}$ of

the wave in Graph 6, (xii) v , E_p , E_k and E_T at point 7, (xiii) position where kinetic energy equals to potential energy, (xiv) time where kinetic energy equals to potential energy, and (xv) position where energy is half.



29. Construct the equation of displacement from the Graph 7 which illustrates the displacement vs. time graph for a mass spring system oscillating in simple harmonic motion. The mass of the system is 500g. Calculate the (i) potential energy of the system at $t = 0.5$ sec, (ii) kinetic energy at $t=2$ sec, (iii) $V_{\max}$, $a_{\max}$, (iv) Total energy at $t=0.6$ sec, and (v) acceleration at $t=0.8$ s.



30. Velocity vs. time graph for a mass spring system oscillating in Simple Harmonic Oscillation (SHO) is illustrated in Graph 8. The mass of the mass-spring system is 500 g. (i) Construct the equation of velocity vs. time, and calculate the (ii) velocity at $t=0.1$ s, (ii) acceleration at $t = 0.2$ s, and (iii) potential energy at $t = 0.3$ s, (iv) kinetic energy at $t=0.4$ s, (v) Construct the equation of displacement vs. time, and also calculate the displacement at $t=0.5$ s, (vi) Construct the equation of acceleration vs. time, and also calculate the acceleration at $t=0.6$ s, (vii) total energy at $t=0.7$ s, (viii) Calculate the

$V_{\max}$, $a_{\max}$, $x_{\max}$ of the SHO, (ix) position where kinetic energy equals to potential energy, (x) time where kinetic energy equals to potential energy, and (xi) At point 5, what is the value of $V_{\max}$, $a_{\max}$, $x_{\max}$, and (xii) At point 2, $x_{\max}=0$ or $V_{\max}=0$?

